Encrypting volumes with NetApp Volume Encryption

Cloud Manager 3.6

Ben Cammett
June 03, 2019

This PDF was generated from https://docs.netapp.com/us-en/occm36/task_encrypting_volumes.html on December 03, 2020. Always check docs.netapp.com for the latest.

Table of Contents

Encrypting volumes with NetApp Volume Encryption 1

Encrypting volumes with NetApp Volume Encryption

NetApp Volume Encryption (NVE) is a software-based technology for encrypting data at rest one volume at a time. Data, Snapshot copies, and metadata are encrypted. Access to the data is given by a unique XTS-AES-256 key, one per volume.

About this task

At this time, Cloud Volumes ONTAP supports NetApp Volume Encryption with an external key management server. An Onboard Key Manager is not supported.

You need to set up NetApp Volume Encryption from the ONTAP CLI. You can then use either the CLI or System Manager to enable encryption on specific volumes. Cloud Manager does not support NetApp Volume Encryption from its user interface and from its APIs.

Learn more about supported encryption technologies.

Steps

1. Review the list of supported key managers in the NetApp Interoperability Matrix Tool.

   Search for the **Key Managers** solution.

2. Connect to the Cloud Volumes ONTAP CLI.
3. Install a NetApp Volume Encryption license on the Cloud Volumes ONTAP system.

   ONTAP 9 NetApp Encryption Power Guide: Installing the license

4. Install SSL certificates and connect to the external key management servers.

   ONTAP 9 NetApp Encryption Power Guide: Configuring external key management

5. Create a new encrypted volume or convert an existing unencrypted volume using either the CLI or System Manager.
   - CLI:
     - For new volumes, use the **volume create** command with the -encrypt parameter.

       ONTAP 9 NetApp Encryption Power Guide: Enabling encryption on a new volume

     - For existing volumes, use the **volume encryption conversion start** command.

       ONTAP 9 NetApp Encryption Power Guide: Enabling encryption on an existing volume with the volume encryption conversion start command

- System Manager:
  - For new volumes, click **Storage** > **Volumes** > **Create** > **Create FlexVol** and then select **Encrypted**.

    ONTAP 9 Cluster Management using System Manager: Creating FlexVol volumes

  - For existing volumes, select the volume, click **Edit**, and then select **Encrypted**.

    ONTAP 9 Cluster Management using System Manager: Editing volume properties

Copyright Information

Copyright © 2020 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.